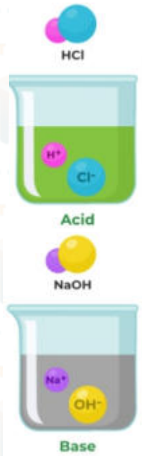
UNIT 08

ACIDS, BASES & SALTS

Arrhenius Theory of Acids and Bases:

The Arrhenius theory of acids and bases was proposed by Swedish chemist Svante Arrhenius in 1884. According to this theory, an acid is a substance that dissociates in water to produce hydrogen ions (H^+), and a base is a substance that dissociates in water to produce hydroxide ions (OH^-). This theory is specifically applicable to aqueous solutions.

Main postulates of the Arrhenius theory of acids and bases:



1. Acid: An acid is a substance that ionizes in water to release hydrogen ions (H^+). The higher the concentration of H^+ ions, the stronger the acid.

$$HA_{(acid)} \rightleftharpoons H^+_{(aq)} + A^-_{(aq)}$$

Examples: HCl, H_2SO_4, CH_3COOH.

2. Base: A base is a substance that ionizes in water to release hydroxide ions (Oh^-). The higher the concentration of OH^- ions, the stronger the base.

$$BOH_{(base)} \rightleftharpoons B^+_{(aq)} + OH^-_{(aq)}$$

Examples: $NaOH$, KOH, $Ca(OH)_2$.

3. Neutralization: When an acid reacts with a base, they undergo a neutralization reaction, resulting in the formation of water and a salt.

$$HCl_{(aq)} + NaOH_{(aq)} \longrightarrow H_2O_{(l)} + NaCl_{(aq)}$$

Limitations:

1. Hydrogen ions in water solutions combine with water to form hydronium ions.
2. The Arrhenius theory doesn't account for the basicity of ammonia or the acidity of carbon dioxide and similar compounds.
3. It is only applicable to reactions that take place in aqueous solutions.

Bronsted Lowry Theory of Acids and Bases:

The Bronsted-Lowry theory of acids and bases, proposed by Danish chemist Johannes Brønsted and English chemist Thomas. It defines acids and bases based on their ability to donate or accept protons (H^+ ions) in chemical reactions. According to this theory:

1. Brønsted Acid: A Bronsted acid is a substance that can donate a proton (H^+ ion) to another species in a chemical reaction. **OR** acid donates its $H+$ ion to its conjugate base.

$$\underset{\text{acid}}{HA} + \underset{\text{base}}{B} \rightleftharpoons \underset{\text{conjugate base}}{A^-} + \underset{\text{conjugate acid}}{HB^+}$$

Example reaction of a Brønsted acid donating a proton:

$$HCl_{(acid)} + H_2O_{(base)} \longrightarrow Cl^-_{(conjugate\ base)} + H_3O^+_{(conjugate\ acid)}$$

In this reaction, HCl donates a proton to H_2O, forming the chloride ion (Cl^-) as its conjugate base and a hydronium ion (H_3O^+) as its conjugate acid.

Accepts proton

$$HCl_{(aq.)} + H_2O \rightleftharpoons Cl^-_{(aq.)} + H_3O^+_{(aq.)}$$

Donates proton

2. Brønsted Base: A Bronsted base is a substance that can accept a proton (H^+ ion) from another species in a chemical reaction.

Example reaction of a Brønsted base accepting a proton:

NH3 (base) + H2O (acid) ⇌ NH4+ (conjugate acid) + OH- (conjugate base) In this reaction, NH_3 accepts a proton from H_2O, forming the ammonium ion (NH_4^+) as its conjugate acid and a hydroxide ion (OH^-) as its conjugate base.

Accepts proton

$$H_2O + NH_{3(aq.)} \rightleftharpoons NH^+_{4(aq.)} + OH^-_{(aq.)}$$

Ammonia

Donates proton

3. Conjugate Acid-Base Pairs: In any Bronsted acid-base reaction, there will be the formation of a conjugate acid and a conjugate base.

Example of a conjugate acid-base pair:

$$CH_3COOH_{\text{(acid)}} + H_2O_{\text{(base)}} \rightleftharpoons H_3O+_{\text{(conjugate acid)}} + CH_3COO-_{\text{(conjugate base)}}$$

Limitations:

1. It does not explain the acidic properties of substances that do not contain hydrogen atoms or H+ ions. For example, boron trifluoride (Bf_3) and aluminum chloride ($AlCl_3$) are strong acids, but it does not contain any hydrogen atoms.

2. It does not explain the reactions between acid oxides and basic oxides. For example, carbon dioxide (CO_2) can react with sodium hydroxide (NaOH) to form sodium carbonate (Na_2CO_3), but this reaction does not involve the transfer of protons.

Lewis Theory of Acid-Base:

The Lewis theory of acids and bases was proposed by American chemist Gilbert N. Lewis in 1923. The Lewis theory is based on electron pair donation and acceptance.

Key points of the Lewis theory of acids and bases:

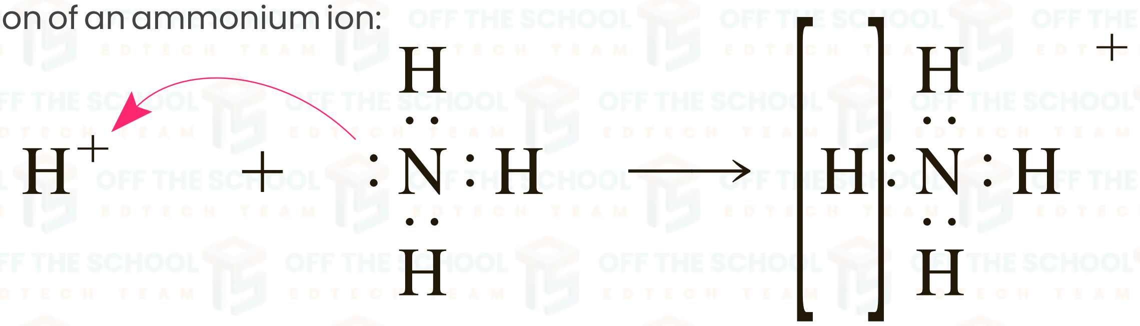
1. Lewis Acid: A Lewis acid is a species that can accept a pair of electrons. It is an electron-pair acceptor.

2. Lewis Base: A Lewis base is a species that can donate a pair of electrons. It is an electron-pair donor.

3. Acid-Base Reaction: In the Lewis theory, an acid-base reaction involves the donation of an electron pair from the Lewis base to the Lewis acid, forming a coordinate covalent bond between the two species.

Example of a Lewis acid-base reaction:

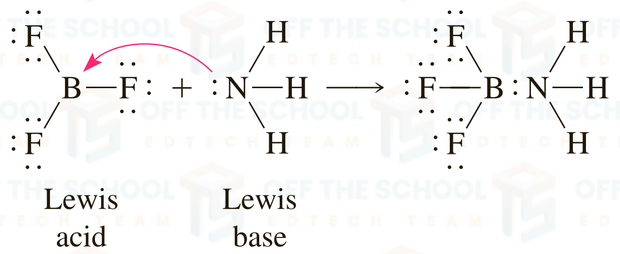
Formation of an ammonium ion:



electron-pair acceptor   electron-pair donor

Formation of a coordination complex between fluorine anion (F^-) and boron trifluoride (Bf_3):

$$BF_{3\ (\text{Lewis acid})} + NH_{3\ (\text{Lewis base})} \longrightarrow [BF_3 \rightarrow NH_3]_{(\text{Lewis acid-base adduct})}$$



Limitations:

1. Limited to electron-pair donation and acceptance: The Lewis theory focuses on the donation and acceptance of electron pairs. An acid is a substance that accepts an electron pair, and a base is a substance that donates an electron pair. However, it doesn't consider other ways substances can react with each other.

2. Doesn't explain reactions involving protons: The Lewis theory doesn't provide a complete explanation for reactions involving protons (H+ ions), which are important in acid-base chemistry. It doesn't account for the transfer of protons between substances.

3. Doesn't consider solvent effects: The Lewis theory doesn't consider the role of solvents, such as water, in acid-base reactions. It doesn't explain how the presence of a solvent can influence the behavior of acids and bases.

CONCEPT OF PH AND POH:

Auto-Ionization Water Ionization:

Water ionization is the process by which water molecules dissociate or break apart into ions in an aqueous solution. In water, a small fraction of water molecules naturally undergo ionization to form hydronium ions (H_3O^+) and hydroxide ions (Oh^-).

The ionization of water is represented by the following chemical equation:

The equilibrium constant of (Kc) of water is given by,

$$K_c = \frac{[H^+][OH^-]}{[H_2O]}$$

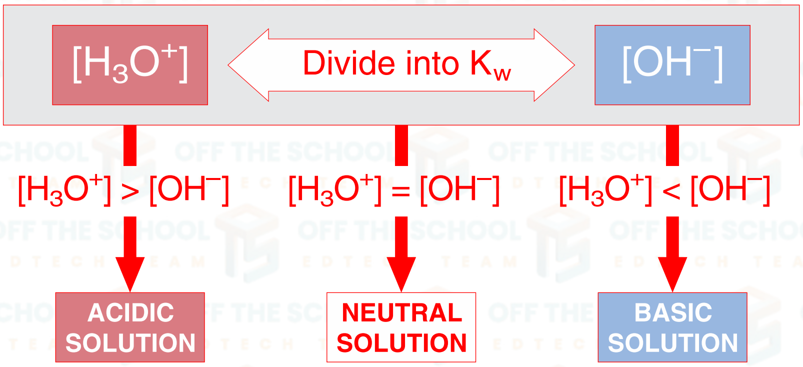
The square brackets represent the molar concentration of species and its units are mole dm. As we know that ionization of water is very small so the concentration is approximately unchanged and consider as constant (KW) so, the equation will be:

$$K_c\,[H_2O] = [H^+]\,[OH^-]$$

$$K_c\,[H_2O] = K_w$$

$$K_w = [H^+][OH^-]$$

Where Kw is ionic product constant of water and its value is ; 1.00×10^{-14} $(mol/dm^2)^2$ at room temperature 25°C.



The relationship between $[H_3O^+]$ and $[OH^-]$ and the relative acidity of solutions.

DEFINITION OF PH AND POH:

pH: pH is a measure of the acidity or alkalinity (basicity) of a solution. It quantifies the concentration of hydrogen ions (H^+) present in a solution.
Mathematically, pH is defined as:

$$pH = -\log[H^+]$$

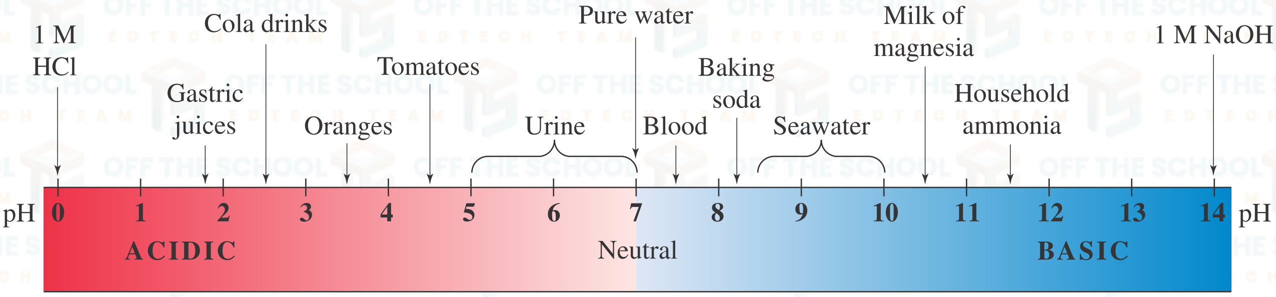
Where $[H^+]$ represents the concentration of hydrogen ions in moles per litre (mol/L).

pOH: pOH is a measure of the hydroxide ion (OH^-) concentration in a solution. It is the negative logarithm of the hydroxide ion concentration.
Mathematically, pOH is defined as:

$$pOH = -\log[OH^-]$$

Where $[OH^-]$ represents the concentration of hydroxide ions in moles per litre (mol/L).

pH Scale: The Danish biochemist S. P. L. Sorensen devised the pH scale while working on the brewing of beer. The pH scale is a numerical scale used to specify the acidity or basicity (alkalinity) of a solution. It quantifies the concentration of hydrogen ions (H^+) in a solution and ranges from 0 to 14.



Salts:

Salt is an ionic compound that contains a cation (from base) and an anion (from acid).

It is present in large quantities in seawater, where it is the main mineral constituent. Salt is important for animal life and saltiness is one of the basic human tastes. Salt is an ionic compound that has a cation other than H and an anion other than OH and is obtained along with water in the neutralization reaction between acids and bases.

Examples: $NaCl$, $CuCl_2$, etc.

$$\textbf{Acid + Base} \longrightarrow \textbf{Salt + water}$$

$$NaOH + HCl \longrightarrow NaCl + H_2O$$

$$2KOH + H_2SO_4 \longrightarrow K_2SO_4 + 2H_2O$$

$$Ca(OH)_2 + 2HNO_3 \longrightarrow Ca(NO_3)_2 + 2H_2O$$

Preparation of Salts:

1. Salts are produced by the action of acids on metals, metal oxides, metal hydroxide, metal carbonates and metal bicarbonates:

$$CaO + H_2SO_4 \longrightarrow CaSO_4 + H_2O$$
$$KOH + HNO_3 \longrightarrow KNO_3 + H_2O$$
$$BaCO_3 + 2HCl \longrightarrow BaCl_2 + CO_2 + H_2O$$

2. Salts are produced by the action of a base with an acid or a metal with a base:

$$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$$
$$HCl + Zn \longrightarrow ZnCl_2 + H_2$$

Types of Salts:

1. Simple Salts: These are salts formed by the direct combination of a metal cation with a non-metal anion.
Examples: Sodium chloride (NaCl), potassium nitrate (KNO_3), calcium sulphate ($CaSO_4$).

2. Acid Salts: Also known as hydrogen salts, acid salts are formed when an acid partially replaces the hydrogen ions with a metal cation.
Examples: Sodium dihydrogen phosphate (NaH_3PO_4), potassium hydrogen sulphate ($KHSO_4$).

3. Basic Salts: Basic salts are formed when a metal hydroxide reacts with an acid, resulting in the formation of a salt and water.
Examples: basic lead carbonate ($PbCO_3 \cdot Pb(OH)_2$), basic copper sulfate ($CuSO_4 \cdot 3Cu(OH)_3$).

Uses of Salts:

1. Food preservation and flavoring in cooking.
2. Water softening to remove hardness in water.
3. Fertilizers for improving crop growth and yield.
4. Pharmaceutical applications in medicine and health.
5. Chemical industry as raw materials and reagents.

CONCEPT OF BUFFER:

A buffer is a solution that resists changes in pH when small amounts of acid or base are added to it. It helps to maintain the pH of a solution relatively stable, even with the addition of acidic or basic substances.

TYPES OF BUFFERS:

1. Acidic buffers:

Acidic buffers have a pH below 7 and consist of a weak acid plus its salt with a strong base.
Examples: A common example is a mixture of acetic acid (CH_3COOH) and sodium acetate (CH_3COONa). This buffer can maintain a pH around 4.75, which is the pKa value of acetic acid.

2. Basic buffers:

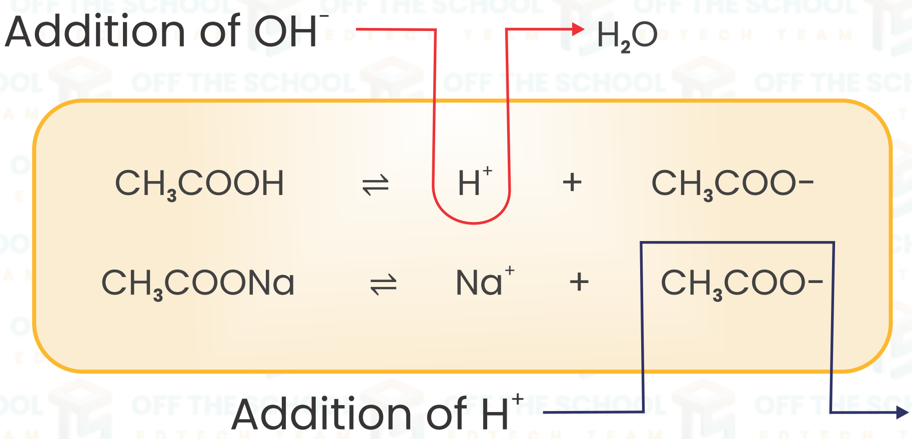
Basic buffers have a pH above 7 and consist of a weak base plus its salt with a strong acid.
Examples: An example is a mixture of ammonia (NH_3) and ammonium chloride (NH_4Cl). This buffer system can maintain a pH around 9.25, which corresponds to the pKa value of the ammonium ion (Nh_4^+).

Buffer action:

Buffer action is the process by which a buffer solution maintains a stable pH despite the addition of small amounts of acid or base. This stability is due to the presence of a weak acid and its conjugate base, or a weak base and its conjugate acid, which can neutralize added acids or bases. The effectiveness of a buffer is limited by its capacity, which is determined by the concentrations of its components and their pH relative to their pKa or pKb values.

Mechanism of Buffering Action:

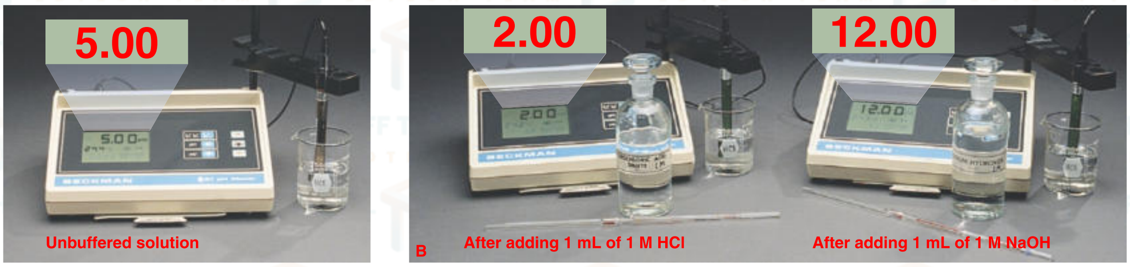
To understand the mechanism of buffer action, we can take the example of an acidic buffer that is made up of a weak acid like acetic acid and its sodium salt, sodium acetate. In this acidic buffer, the solution contains equimolar amounts of acetic acid and sodium acetate. Usually, a large number of sodium ions (Na^+), acetate ions (CH_3COO^-) and undissociated acetic acid molecules are present. The salt exists completely as ions.



Here, the buffer will consist of both acid (CH_3COOH) and its conjugate base (CH_3COO^-). If we add a small quantity of acid, the hydrogen ions will be removed by the conjugate base (CH_3COO^-). It is represented as follows:

$$H^+_{(aq)} + CH_3COO^-_{(aq)} \longrightarrow CH_3COOH_{(aq)}$$

Here, the ethanoic acid will only be slightly dissociated in the form of CH_3COOH, which means that it will not contribute any H+ ions. Therefore, the pH of the resulting solution will remain more or less constant. The added H+ ions are also removed, due to which there is no appreciable decrease in pH.



The effect of adding acid or base to an unbuffered solution. **A,** A 100-mL sample of dilute HCl is adjusted to pH 5.00. **B,** Adding 1 mL of strong acid *(left)* or of strong base *(right)* changes the pH by several units.

The effect of adding acid or base to a buffered solution. **A,** A 100-mL sample of an acetate buffer (1 *M* CH_3COOH mixed with 1 *M* CH_3COONa) is adjusted to pH 5.00. **B,** Adding 1 mL of strong acid *(left)* or of strong base *(right)* changes the pH negligibly.

Applications of buffer solution:

Buffer solutions are essential in various fields due to their ability to maintain stable pH levels:

Biochemistry: They regulate blood pH around 7.3 to 7.4 using bicarbonate and carbonic acid buffers.

Industrial Processes: Utilized in fermentation, dyeing, and pharmaceutical manufacturing to ensure optimal pH for reactions.

Agriculture: Used to adjust soil pH to improve crop yields.

Analytical and Pathological Labs: Critical in experiments and tests that require precise pH conditions.

Food Industry: Important for preserving the flavour, appearance, and microbiological stability of food products.

LEVELING EFFECT:

The leveling effect is the process by which strong acids or bases appear equally strong in water due to the solvent's tendency to limit the strength of the acids or bases to that of the native ions of the solvent (H_3O+ for acids and $OH-$ for bases).

Example:

An example of the leveling effect is when hydrochloric acid (HCl) and nitric acid (HNO_3) are both dissolved in water. Although they may have different strengths as pure compounds, in aqueous solution, they both fully ionize to produce hydronium ions (H_3O+). Thus, despite their intrinsic differences in acid strength, they exhibit the same acidity level in water.

For hydrochloric acid:

$$HCl_{(aq)} + H_2O_{(l)} \longrightarrow H_3O_{(aq)} + Cl^-_{(aq)}$$

IMPORTANT QUESTIONS

1. Define pH & pOH of a solution? Also show that pH + pOH = 14.

pH is a measure of how acidic or basic a water-based solution is. It stands for 'potential of Hydrogen' or 'power of Hydrogen'. The pH scale ranges from 0 to 14:

- A pH less than 7 is acidic
- A pH of 7 is neutral
- A pH greater than 7 is basic (or alkaline).

pOH is similar to pH but measures the concentration of hydroxide ions (OH^-). The relationship between pH and pOH is given by the equation: pH + pOH = 14 This is true at 25°C because the product of the concentrations of H^+ and OH^- ions in water (the ion product constant for water, Kw) is always $1x10^{-14}$ at this temperature.

2. Why the aqueous solution of NH_4Cl is acidic and Na_2CO_3 is alkaline.

NH_4Cl is acidic because it forms NH_4^+ and Cl^- ions in water. NH_4^+ can give away a H^+ ion to become NH_3, making the solution acidic.

Na_2CO_3 is alkaline because it forms Na^+ and CO_3^{2-} ions in water. CO_3^{2-} can take up H^+ ions to form HCO_3^-, reducing the concentration of H^+ ions and thus the solution becomes more basic.

3. Write down conjugate base of each of the following acids:

H_2SO_4, H_2S, NH_4^+, HCOOH

- The conjugate base of **H_2SO_4 (sulfuric acid)** is HSO_4^- (hydrogen sulfate).
- The conjugate base of **H_2S (hydrosulfuric acid)** is HS^- (hydrogen sulfide).
- The conjugate base of **NH_4^+ (ammonium ion)** is NH_3 (ammonia).
- The conjugate base of **HCOOH (formic acid)** is $HCOO^-$ (formate ion).

4. What is meant by self-ionization of water? Write the expression of Kw. What is its value at 25°C?

The self-ionization of water refers to the reaction where water molecules react with each other to form hydronium ions (H_3O^+) and hydroxide ions (OH^-).

The expression for the ion-product constant of water (Kw) is: $K_w = [H_3O^+][OH^-]$

At 25°C, the value of Kw is 1×10^{-14}

5. Write equation and indicate the conjugate acid-base pairs for the following:

(i) Acetic acid & ammonia **(ii)** Hydrochloric acid & water

$$CH_3COOH + NH_3 \rightleftharpoons CH_3COO^- + NH_4^+$$

- The conjugate base of acetic acid is acetate (CH_3COO^-).
- The conjugate acid of ammonia is ammonium (NH_4^+).

$$HCl + H_2O \rightleftharpoons Cl^- + H_3O^+$$

- The conjugate base of hydrochloric acid is chloride (Cl^-).
- The conjugate acid of water is hydronium (H_3O^+).

6. Explain Brønsted-Lowry theory of acids and bases. What is Meant by conjugate acid-base pair give examples?

Brønsted-Lowry Theory defines an acid as a substance that can donate a proton (H^+), and a base as a substance that can accept a proton. A conjugate acid-base pair consists of two substances that transform into each other by the loss or gain of a proton.

Examples:

When hydrochloric acid (HCl) donates a proton to water (H_2O), it becomes the chloride ion (Cl^-). Here, HCl is the acid, and Cl^- is its conjugate base.

Ammonia (NH_3) can accept a proton to become ammonium (NH_4^+). In this case, NH_3 is the base, and NH_4^+ is its conjugate acid.

7. Define the Process of Hydrolysis. Explain the Behavior of Each of the Following Salts in Aqueous Solution.

(i) K_2CO_3, **(ii)** $(NH_4)_2SO_4$, **(iii)** $NaNO_3$.

Hydrolysis is a reaction where a salt reacts with water, leading to the formation of an acid and a base:

K_2CO_3 (Potassium Carbonate): This salt comes from a strong base (KOH) and a weak acid (H_2CO_3), so it undergoes basic hydrolysis. The solution becomes alkaline because CO_3^{2-} can remove H^+ from water, increasing OH^- concentration.

$(NH_4)_2SO_4$ (Ammonium Sulfate): This salt comes from a weak base (NH_3) and a strong acid (H_2SO_4), so it undergoes acidic hydrolysis. The solution becomes acidic because NH_4^+ can donate a proton to water, increasing H_3O^+ concentration.

$NaNO_3$ (Sodium Nitrate): This salt is from a strong base (NaOH) and a strong acid (HNO_3). It does not undergo significant hydrolysis, so the solution remains neutral.